

Brian Glasheen

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Summary

I am passionate about all things computer, transitioning my career into computer graphics and real-time rendering. I'm looking to apply my experience to rendering, simulation, and visual computing while learning along the way.

Work History

Associate Software Engineer, L3Harris Technologies, TX

January 2025 - Current

- Developed C++ benchmarking suite from scratch to validate performance critical mission systems that guaranteed compliance with strict requirements
- Designed and implemented communication interfaces between distributed systems with Google Protocol Buffers, using data driven approaches and optimized data structures to meet high throughput demands
- Developed and maintained GitLab CI/CD pipelines with CMake and Gradle to streamline application testing and deployment, improving reliability and reducing deliverable release time
- Deployed and managed software on Kubernetes clusters to ensure scalable, reliable, and efficient delivery of compute, storage, and memory intensive applications

Software Engineer Intern, StandardAero, TX

June 2024 - September 2024

- Developed and deployed a full-stack form automation tool integrating data from multiple cross functional teams to reduce manual labor requirement by 90% and minimize data entry errors
- Designed and integrated a real-time visualization dashboard combining multiple data sources, enabling real-time trend detection in aircraft part usage, improving inventory management and part over/under order

Teaching Assistant, Texas A&M College of Engineering, TX

August 2022 - December 2024

- Assisted professors in teaching and grading engineering coding classes, providing guidance to students on debugging, algorithm design, profiling, and best practices
- Led weekly physics labs for students, presented briefings on current experiments, ensured smooth execution, and provided assistance to students
- Coordinated with fellow peer teachers to manage deadlines and oversee multiple classrooms of ~100 students, ensuring consistent instruction and timely grading

Projects

Vulkan renderer - github.com/BrianGlasheen/fireball

- Designed and implemented a GPU-driven, bindless Vulkan renderer built around an ECS (Entity Component System) architecture, using VkBootstrap for Vulkan initialization and Vulkan Memory Allocator (VMA) for efficient GPU memory management
- Integrated meshoptimizer to generate automatic LODs (Levels of Detail), dynamically selected during GPU culling to improve rendering performance
- Developed physically based rendering (PBR) GLSL shaders, compiled to SPIR-V via CMake, implementing a realistic lighting model

Glow, Modern OpenGL renderer & game framework - github.com/BrianGlasheen/glowing-engine

- Implemented GPU frustum and occlusion culling, moment-based order-independent-transparency, toggleable bindless textures, physically based rendering techniques, GPU & CPU skeletal animation, compute shader skinning, Jolt physics integration, multi-channel signed distance field text rendering, clustered forward+ lighting, parallel-split shadow maps, scene (de)serialization in .yaml
- Used built-in profiler and RenderDoc for debugging and performance analysis during development

Software Raytracer - github.com/BrianGlasheen/raytracing

- Implemented a CPU recursive Monte Carlo path tracer supporting reflective and refractive materials using cosine-weighted hemisphere sampling, Schlick's approximation, Russian roulette path termination, and importance sampling
- Implemented bottom level and top level acceleration structures via a bounding volume hierarchy for accelerated scene queries

Education & Certifications

- B.S. in Computer Science from Texas A&M University
- CompTIA Security+

August 2021 - December 2024